DS 3010

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Module 3: Model Selection

Part 2: Step Wise Selection Methods

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Best Subset Selection (p <30)

- This is an exhaustive search. It will produce an optimal model based on some criteria.
- Computationally infeasible for values of p greater than around 40, even with extremely fast modern computers.
- Only work for least squares linear regression

Model Selection Strategy

P is small (p < 30)

- Best subset selection

P is large (p >= 30)

- Forward selection
- Backward selection
- Hybrid approach

Stepwise Selection

- For computational reasons, best subset selection cannot be applied when p is large.
- We need an alternative: **stepwise selection**
- These methods are based on greedy algorithms.
- Greedy here means we make the choice of 'best' model based on our current information, but this is not necessarily the 'best' global choice.
- That means, unlike subset selection, we do not consider all possible models.
- Therefore, greedy decisions may lead to a **suboptimal** choice of model.

Forward Stepwise Selection

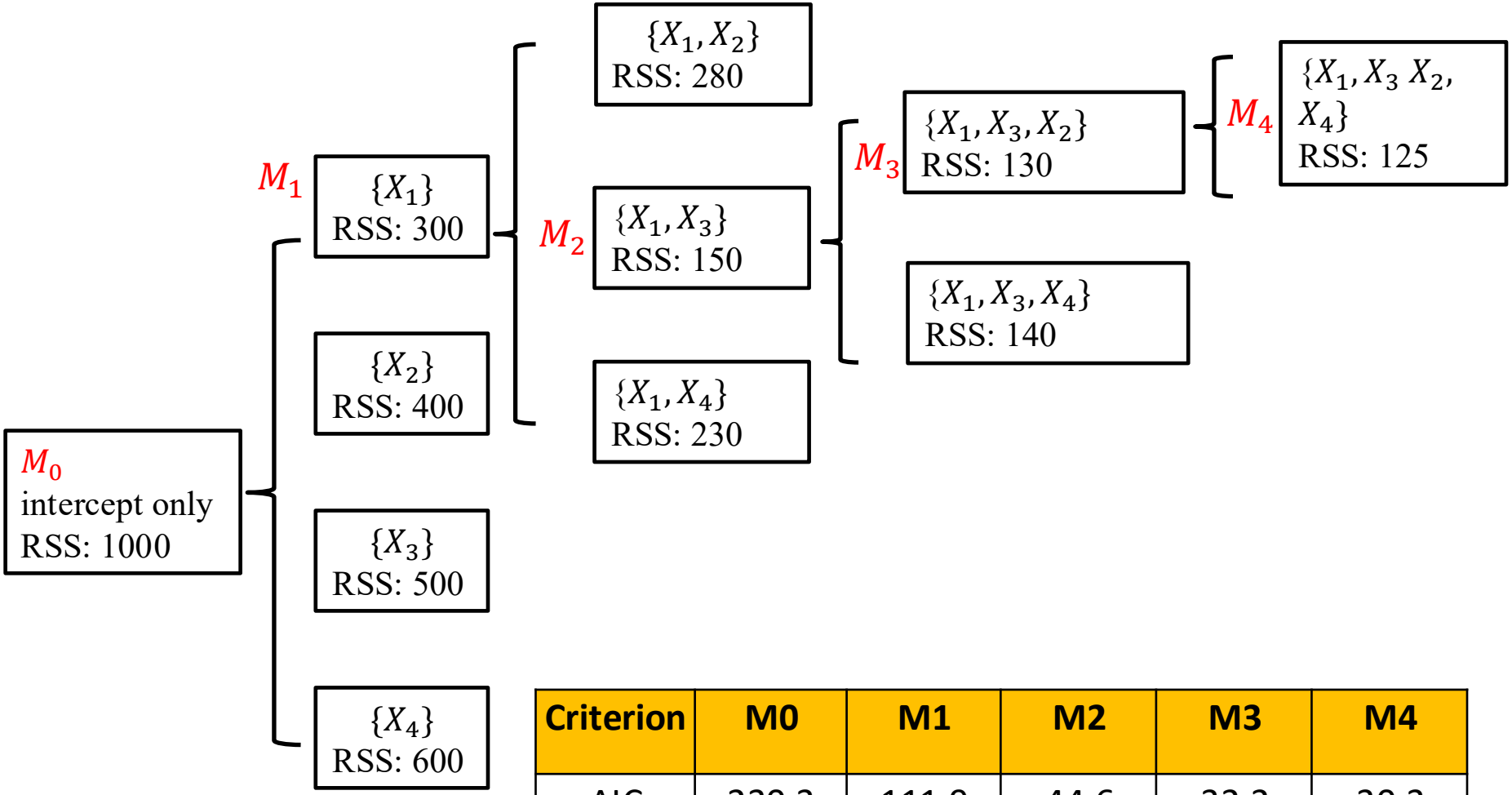
Algorithm:

1. Let M_0 denote the null model, which contains no predictors.

2. For $k = 0, \ldots, p - 1$
   a. Consider all $p - k$ models that augment the predictors in M_k with one additional predictor.
   b. Choose the best among these $p - k$ models and call it M_{k+1}. Here best is defined as the model with the smallest RSS.

3. Select a best model $M_1, \ldots, M_p$ using cross-validated error, AIC, BIC, Mallow's C_p, or adjusted R^2.

Forward Stepwise Selection



Criterion	M0	M1	M2	M3	M4
AIC	230.3	111.9	44.6	32.2	30.3
BIC	230.3	114.8	49	40.0	40.7
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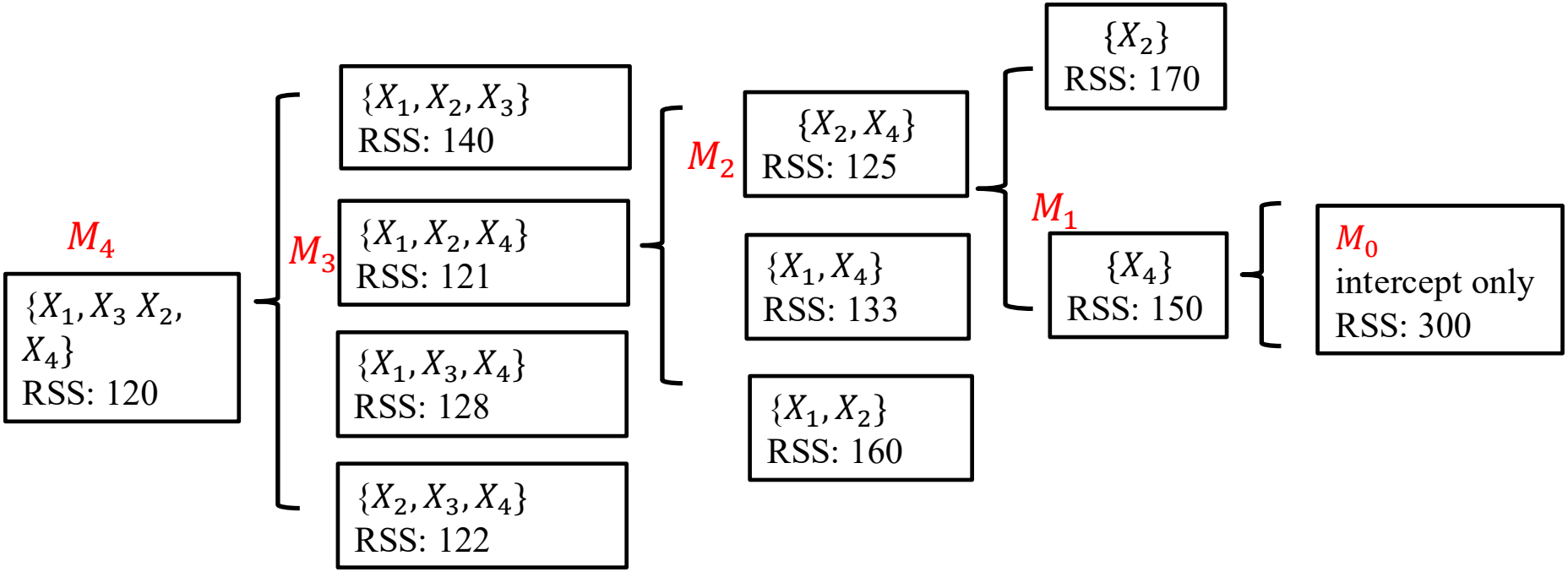
Backward Stepwise Selection

Same algorithm as forward selection, but now direction moves backward.
Initial model: full model
Iteratively remove each predictor one at a time.

Backward Stepwise Selection



Criterion	M0	M1	M2	M3	M4
AIC	109.9	42.6	26.3	25.1	26.2
BIC	109.9	45.2	31.5	32.9	36.6
…					

Forward vs. backward

- If your goal is prediction and a more complex model is acceptable: backward selection.
- If, all things considered, you would prefer a simpler model: forward selection.
- If you have high-dimensional data: forward selection

Hybrid approach

- Best subset, forward, and backward stepwise will generally give similar but not identical models.
- An alternative is a hybrid version of forward/backward selection: stepwise selection.
- At each step, you are not restricted to one direction. You can add and remove predictors.
- Stepwise selection attempts to mimic subset selection while still retaining computational advantage.
- See R script: stepwise.R